

THE TALE OF TWO AUTONOMOUS CITIES

How agents, skills, tools and MCPs turn a human objective
into coordinated digital work

*“The visible answer is only the final page of an invisible
organization.”*

ALEJANDRO REYNOSO

JUDGE BUSINESS SCHOOL.UNIVERSITY OF CAMBRIDGE.JULY 2026

The Central Idea

The architecture in one sentence

An autonomous system is not a single all-knowing machine. It is a coordinated organization in which an agent reasons, skills shape the method, tools make actions possible, MCPs create standardized connections, external systems execute specialized operations, and the human retains authority over purpose and judgment.

Two cities, two kinds of work

The first city produces a daily market report. It demonstrates how an autonomous system plans, retrieves evidence from several sources, normalizes the results and synthesizes them into analysis.

The second city manages the morning inbox of a bank chief executive. It demonstrates a more demanding form of autonomy: the system reads, classifies, replies, consults supporting applications, escalates sensitive matters and prepares a decision briefing.



Introduction: Two Cities You Never See

There is a curious paradox at the heart of modern artificial intelligence. From the user's perspective, the experience is often astonishingly simple. A person states an objective, perhaps in a single sentence, and a polished result appears. A report is prepared. An inbox is organized. Meetings are checked. Evidence is gathered. Replies are drafted. Decisions are highlighted.

The simplicity is real, but it is only the visible surface.

Beneath it lies an elaborate digital organization: a network of reasoning components, specialized capabilities, connectors, applications, permissions and external services. These components do not all perform the same function. Some decide. Some translate. Some retrieve. Some execute. Some verify. Some refuse to proceed without authorization. Together they create the impression of a coherent autonomous worker.

The most useful way to understand this architecture is not to imagine a single artificial brain that knows and does everything. It is to imagine a city.

A city is intelligent not because every citizen possesses every skill, but because roles are differentiated and coordinated. The architect does not pour every foundation. The physician does not manufacture every instrument. The mayor does not personally operate every public service. Intelligence emerges from specialization, communication, institutions and governance.

Autonomous systems work in much the same way.

In this paper, we enter two such cities. The first is the Research City of an investment bank, where a deputy agent must assemble the evidence and reasoning required for a daily market report. The second is the Executive City surrounding the chief executive, where an autonomous assistant confronts the daily flood of email and converts it into a small number of decisions that truly require leadership.

The stories are deliberately human and generous in detail because technical language often hides the essential choreography. The objective here is to make the choreography visible.

The conceptual map

The human states the purpose and retains authority.

The agent interprets the objective, plans and selects actions.

Skills provide reusable methods for performing recurring kinds of work.

Tools are the concrete actions the agent is permitted to invoke.

MCP servers or connectors expose those tools in a standard, machine-readable way and mediate access to applications.

External applications store data or execute operations.

Governance determines what may happen automatically and what requires approval.

1 The First Autonomous City: The Research Department

The sun has barely risen over the financial district. The glass towers are still reflecting the pale blue of the morning, but the research floors of the bank are already awake. Traders in Asia have completed much of their day. Europe is opening. New York is preparing for the next wave of decisions.

Every morning, clients and senior executives expect the bank's market report. They do not merely want a table of prices. They want a coherent explanation: What moved? Why did it move? Which developments matter? What should investors watch next?

Alex, the Head of Research, enters his office with a cup of coffee. He does not begin by searching websites. He does not open six terminals or dictate a list of data providers. He calls his deputy, Daniel.

The mission

Alex: "Prepare today's global market report. Tell me what happened yesterday, explain why markets moved, and identify what investors should understand before trading begins."

This is a statement of purpose, not a technical specification. Alex defines the destination without prescribing every road.

1.1 The deputy does not search first; he plans

Daniel closes the door and thinks. To explain the markets, he will need evidence from several domains. Equity prices alone will not be enough. He needs sovereign yields, exchange rates, commodities, corporate news, economic releases and perhaps internal research written by the bank's specialists.

No data has yet been retrieved. This distinction matters. Planning is not the same as execution.

In an autonomous system, Daniel represents the agent: the component that interprets the objective, decomposes it into subtasks and determines an order of operations. The agent may decide that some tasks can run in parallel, while others depend on earlier results. It may also revise the plan if the evidence points in an unexpected direction. This combination of model-based reasoning, instructions and tool use is a common foundation for practical agent design [1].

Daniel writes a working plan:

1. Retrieve the principal market movements.
2. Retrieve the macroeconomic releases and policy developments.
3. Search relevant company and geopolitical news.

4. consult the bank's internal research archive.
5. Compare and reconcile the evidence.
6. Produce a narrative with conclusions, risks and watch points.

The plan is the first expression of intelligence in the process. It transforms a broad request into a sequence of achievable operations.

1.2 The marketplace of capabilities

Daniel does not possess direct knowledge of every technical interface. Instead, he has access to a catalog of available capabilities. One office can retrieve market prices. Another can obtain government economic data. Another can search the internal knowledge base. Another can read current news.

In technical terms, these capabilities may be exposed by MCP servers, plugins or application-specific connectors. Each advertises the tools it offers, their descriptions and the parameters they require.

The agent therefore does not face an undifferentiated universe. It sees a structured menu of possible actions.

For example:

Capability office	Services exposed to the agent
Market Data	Historical prices, current quotes, dividends, splits, volumes
Economic Data	Yields, inflation, employment, policy rates, macro series
News	Search, filtering, article retrieval, event monitoring
Internal Research	Document search, report retrieval, analyst notes, prior briefings

The descriptions matter because they influence routing. If several services can supply prices, the system may be instructed to prefer the licensed institutional source, use a public source as fallback, or select according to cost, freshness, speed or jurisdiction.

The MCP does not normally make this strategic choice. The agent, or a policy layer supervising the agent, selects the appropriate capability.

1.3 A request becomes a structured invocation

Daniel begins with equity prices. He says, in human terms, "I need Apple's daily prices for the last five years."

The software does not simply forward this sentence blindly to Yahoo Finance. The agent maps the request to a particular tool and prepares the required arguments. Behind the conversation, the invocation may resemble:

```
get_historical_prices(
    symbol = "AAPL",
    start_date = "2021-07-23",
    end_date = "2026-07-23",
    interval = "1d",
```

```
    adjusted = true  
)
```

This is the boundary between semantic reasoning and operational execution. The agent understands the meaning of the need; the tool call expresses that need in a precise form.

An important correction to the human analogy

The MCP server is not normally a human analyst listening to an open-ended conversation and deciding what the requester probably meant. It is coded infrastructure. The open-ended interpretation occurs mainly in the agent. The MCP server receives a named tool call with structured parameters, validates it, applies permissions, invokes the provider and returns a result.

1.4 The MCP as a secure service desk

Sophia, the Market Data specialist in our story, represents the MCP service desk. Her office exposes the available market-data tools. It may also hold credentials, enforce rate limits, validate parameters, log requests and translate between the standard tool interface and the proprietary provider interface.

Yahoo Finance does not know that Alex wants a morning report. It receives a narrow operational request: retrieve a defined series for a defined instrument over a defined period.

The provider performs the operation and returns data.

The result may arrive with unfamiliar field names, provider-specific metadata or error codes. The MCP server may convert these into a consistent response schema before returning them to the agent.

This normalization is particularly valuable when several providers represent equivalent concepts differently. One might call a field `Adj Close`; another, `PX_LAST`; another, `ClosingPrice`. A normalized result allows the agent to work with a common vocabulary.

1.5 The city begins to move

The same choreography now unfolds across the city.

The agent requests Treasury yields from the economic-data capability. It searches news for market-moving events. It retrieves internal notes. It may use a calculation tool to compute returns, changes in basis points, volatility or correlations.

Some operations occur in parallel. Others trigger follow-up questions. A sharp move in a bank stock may cause the agent to search regulatory filings or earnings releases. An unexpected change in yields may lead it to inspect central-bank commentary.

The plan is therefore not necessarily rigid. A capable autonomous system observes intermediate results and adapts.

Yet adaptation does not mean unlimited freedom. The system operates within the tools, data sources and permissions it has been given.

1.6 Information becomes insight

When the evidence has been collected, Daniel's role changes. Retrieval gives way to synthesis.

He compares the movements. Technology equities rose as long-term yields declined. Oil fell after an unexpected inventory build. The dollar weakened following softer economic data. Corporate earnings altered sector performance. International developments influenced risk appetite.

The system must distinguish correlation from explanation, direct evidence from inference, and current facts from stale context. A trustworthy design should retain citations, timestamps and source identities so that the final narrative can be audited.

The data are not the report. They are the materials from which the report is built.

Daniel writes, revises and organizes. He identifies the central story, explains competing forces, highlights uncertainty and prepares a watch list for the day ahead.

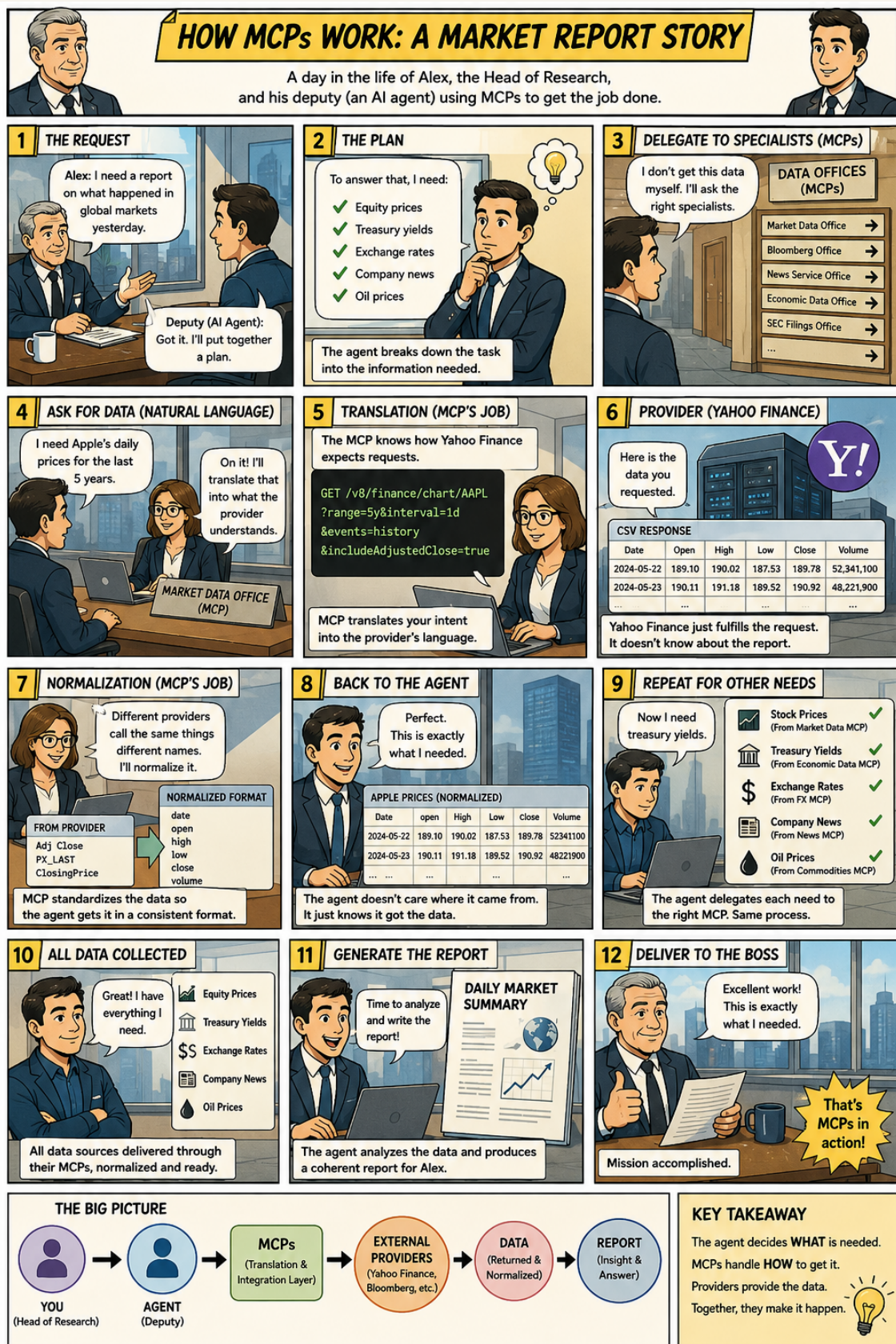
Finally, he returns to Alex's office.

The visible result

Daniel: "The report is complete. Technology shares led the rally as long-term yields declined, while weaker energy prices reflected the inventory surprise. The principal risk today is the policy communication expected this afternoon."

Alex: "Excellent. Clear, evidenced and actionable."

To Alex, the process appears simple. He asked for a report and received one. But the answer is the visible output of an invisible city: planning, routing, authentication, retrieval, normalization, calculation, synthesis and review.



Visual interlude: the Research City in comic form.

2 The Second Autonomous City: The Executive Office

The second city awakens before the chief executive arrives.

It is seven o'clock in the morning. The upper floors of the bank are quiet, but the chief executive's inbox is already crowded. Communications have arrived from directors, regulators, clients, employees, counterparties, advisers, universities, service providers and automated systems.

Each message demands a small portion of attention. Together, they threaten to consume the most scarce resource in the institution: executive judgment.

The autonomous assistant begins with a standing mission:

The executive mandate

Protect the CEO's time. Handle what is routine. Escalate what is consequential. Never confuse speed with authority.

This workflow is more demanding than the market-report process because it does not merely retrieve information. It can alter the external world. It may send messages, move meetings, archive records and create commitments. The governance requirements are therefore substantially higher.

2.1 The morning trigger

The workflow begins because of a schedule: every morning at a defined time, the system is activated. The trigger itself may be implemented by an automation service, an orchestrator or a scheduled job.

The agent then applies a recurring skill: morning inbox triage. A skill is not the email application. It is the method used to perform the task. It may specify the sequence of steps, the classification criteria, the permitted actions, the required safeguards and the format of the final briefing.

A well-designed triage skill might instruct the agent to:

1. Retrieve unread messages from the relevant period.
2. Read threads rather than isolated messages when context matters.
3. Classify each item by urgency, impact, sensitivity and required action.
4. Draft or send routine responses only within approved categories.
5. Consult the calendar, contacts or internal documents when needed.
6. Escalate critical, ambiguous or high-risk communications.

7. Produce a concise executive briefing with recommended actions.

The skill gives the agent a disciplined operating procedure.

2.2 Entering the email application

The assistant does not reach into the mailbox magically. It uses an authorized tool exposed by an email connector or MCP server. The server communicates with Gmail, Outlook or the bank's internal mail platform through the relevant API.

Authentication and authorization are crucial. The system must know whose mailbox it may access, what operations it may perform and under what conditions.

Reading messages is one permission. Sending a reply is another. Deleting messages is another. Accessing attachments may carry additional restrictions.

The MCP or connector provides the operational bridge, but the policy layer determines the boundaries.

2.3 The flood

The inbox contains 126 unread messages.

Some are straightforward: a meeting confirmation, a travel itinerary, an invoice receipt, a newsletter, a thank-you note. Others are subtle. A polite message from a regulator may carry serious implications. A brief note from the chair of the board may require immediate attention. A client complaint may be operationally small but reputationally significant.

The agent reads and classifies.

Classification is not merely keyword matching. It requires context. The sender's identity matters. The relationship matters. The subject matter matters. Deadlines matter. Attachments matter. The conversation history may matter.

The agent may assign categories such as:

Category	Meaning and default treatment
Critical	Requires the CEO's judgment, approval or immediate awareness; no autonomous response unless explicitly authorized
Important	Merits review but may be delegated or scheduled
Routine	Can be handled according to approved templates and policies
Informational	Retain, label or archive; generally no response
Suspicious	Potential security, fraud or compliance issue; isolate and escalate

The agent's classification is an exercise in reasoning. The email MCP does not ordinarily decide whether a message from the board is critical. It provides the email content and the available mail operations. The agent interprets significance.

2.4 Routine does not mean careless

The assistant begins handling approved routine messages.

A meeting invitation can be acknowledged. A student or colleague asking for a document can receive the authorized link. A standard confirmation can be sent. A newsletter can be archived.

But even routine actions should be constrained. The assistant may be allowed to send confirmations and acknowledgments while being prohibited from making financial commitments, expressing legal positions, disclosing confidential information or accepting obligations.

The distinction between drafting and sending is especially important. A low-risk category may permit automatic sending. A slightly more sensitive category may permit only a draft for review.

Autonomy is granular

An autonomous system is not simply “on” or “off.” Different actions can have different authority levels. Reading may be automatic. Classifying may be automatic. Drafting may be automatic. Sending may require approval. Deleting may be prohibited. Commitments above a threshold may always require the CEO.

2.5 Calling the other offices

An email asks to move a meeting. The assistant needs availability, so it invokes a calendar tool. The Calendar MCP translates the request into the calendar application’s interface and returns suitable times.

Another message comes from an unfamiliar name. The assistant invokes the contacts capability to identify the person and understand the relationship.

A third email refers to an earlier board paper. The assistant searches the internal document repository.

The email task has become a multi-application workflow. This is one of the defining features of autonomous systems: they coordinate across systems rather than operating inside a single application silo.

The agent remains the orchestrator. The skills provide the procedure. The tools provide concrete actions. The MCPs or connectors provide standardized access. The applications maintain the underlying records.

2.6 The critical few

After the routine traffic has been processed, five messages remain.

One is from the chair of the board concerning risk exposure. One is from a regulator requesting information. One is from the chief executive of a major client. One proposes a strategically important research partnership. One concerns a contract renewal with material consequences.

The assistant does not reply.

This restraint is not a failure of autonomy. It is evidence of good governance.

The system recognizes that these communications require authority, judgment, institutional context or accountability that belongs to the CEO.

Instead, it prepares a structured briefing for each item:

- a concise summary of the message;
- why it matters;
- the deadline or urgency;
- relevant context from prior communications;
- supporting documents;
- recommended options;
- a proposed response clearly marked as a draft.

2.7 The CEO arrives

At eight-thirty, Alex enters the office. The inbox no longer presents itself as an undifferentiated wall of messages.

The system reports:

The morning briefing

126 messages reviewed.
94 informational items archived or filed.
18 routine responses completed within policy.
9 calendar matters coordinated.
5 critical communications require executive judgment.

The chief executive spends twelve focused minutes on the five consequential items. He modifies two proposed replies, approves one, requests further analysis on another and calls the chair regarding the fifth.

The system has not replaced leadership. It has protected the conditions under which leadership can occur.

BEHIND EVERY GREAT CEO... AN INVISIBLE ARMY OF AGENTS

A MORNING IN THE LIFE OF YOUR AI EXECUTIVE ASSISTANT

YOU RUN THE BANK.
I RUN THE NOISE.
FOCUS ON WHAT MATTERS.
I'LL HANDLE THE REST.
- YOUR AI ASSISTANT

1 7:00 AM - THE MISSION BEGINS

Good morning, CEO. Time to secure your day.

- ✓ Check inbox
- ✓ Classify emails
- ✓ Handle routine replies
- ✓ Flag critical ones
- ✓ Prepare executive briefing

Every morning, I execute your protocol. Relentlessly. Accurately. Quietly.

2 ACCESSING THE GATEWAY

Connecting to Email Services via secure MCP.

EMAIL MCP

Authentication. Authorization. Encrypted. Compliant. Instant.

3 THE FLOOD

INBOX - 126 UNREAD

- ✕ Board of Directors
- ✕ Client - Global Corp.
- ✕ Regulatory Update
- ✕ Meeting Invitations
- ✕ Marketing Newsletter
- ✕ Vendor Invoice
- ✕ Student Loan Question
- ✕ Travel Confirmation
- ✕ Research Collaboration
- ✕ ... and more

The world writes to you. I make sense of it.

4 INTELLIGENT TRIAGE

- ✕ CRITICAL Requires your immediate attention
- ✓ IMPORTANT Needs your review
- ✓ ROUTINE I'll handle
- ✕ NOISE Archive / Ignore

I don't just read emails. I understand them.

5 HANDLING THE ROUTINE

- ✕ Student Extension Request → Replied
- ✕ Meeting Confirmation → Confirmed
- ✕ Invoice Received → Recorded
- ✕ Newsletter: Daily Market Update → Archived
- ✕ Flight Itinerary → Filed

Consider it done. Polite. Professional. In your voice.

I protect your time by eliminating decisions that don't need you.

6 CALLING IN THE SPECIALISTS

Need context. Calling Calendar MCP.

CALENDAR MCP

Also looping in Contacts MCP.

CONTACTS MCP

No single agent knows everything. Together, we bring complete context.

7 GATHERING CONTEXT

CALENDAR RESULTS

Mon	Busy
Tue	Busy
Wed	Available
Thu	Busy

CONTACT INFO

James Whitfield
Former Client
Global Corp.
Relationship: 7 years
Risk Level: Medium

I don't just process emails. I connect the dots.

8 THE CRITICAL FEW

5 ITEMS REQUIRE YOUR ATTENTION

- Board of Directors Q2 Risk Exposure Report
- Global Corp. (CEO) Urgent: Market Volatility Impact
- Regulatory Authority Request for Information
- Research Partner Breakthrough Proposal
- Major Client Contract Renewal Decision

Only what matters. Everything else, handled.

9 EXECUTIVE BRIEFING

MORNING BRIEFING - MAY 21

- Board of Directors - Q2 Risk Exposure
Summary: Request for updated risk metrics before tomorrow's meeting.
Recommended Action: Review attached summary, approve scenario.
Deadline: Tomorrow 9:00 AM **HIGH**
- Global Corp. (CEO) - Volatility Impact
Summary: Concern about FX movement affecting operations.
Recommended Action: Send reassurance & propose call.
Deadline: Today **HIGH**

Summarized. Prioritized. Actionable. So you can lead, not read.

10 YOU DECIDE. I EXECUTE.

Perfect. You've given me clarity and control.

You focus on judgment. I handle the execution.

11 BACK TO WORK

Critical replies drafted.
Meetings scheduled.
Teams notified.
Day protected.

The machine never sleeps. So you can change the world.

12 THE INVISIBLE ARCHITECTURE

YOU (CEO)

AI EXECUTIVE ASSISTANT

EMAIL MCP → Gmail / Outlook / Contacts

CALENDAR MCP → Google Calendar / Exchange

CONTACTS MCP → Contacts DB / CRM

Agents think. MCPs connect. Systems execute. You lead.

THE PROMISE

In a world of endless information, your advantage is not knowing more. It's focusing on what truly matters.

I PROTECT YOUR TIME. YOU PROTECT THE FUTURE.

This is not automation. This is augmentation. This is how leaders win.

Visual interlude: the Executive City in comic form.

3 What the Two Cities Reveal

The two stories appear different. One creates a market report; the other manages executive communications. One is primarily analytical; the other is operational. One retrieves and synthesizes; the other reads, classifies and acts.

Yet their internal architecture is strikingly similar.

3.1 The human defines purpose

In both cities, the process begins with a human objective. The Head of Research requests a market report. The CEO establishes the standing mandate for email triage.

The human does not need to specify every technical step. But the human or institution must define the objective, constraints, authority and standards of success.

3.2 The agent performs semantic work

The agent interprets the request. It decides what information is needed, what actions are appropriate and which available capabilities should be used.

This is where ambiguity is resolved and context is applied. It is also where errors in judgment can arise. Clear tool descriptions, strong instructions and well-designed skills improve the quality of these decisions.

3.3 Skills encode methods

A skill is a reusable operational discipline. The market-report skill might prescribe source coverage, calculations, citation requirements and report structure. The inbox-triage skill might prescribe classification rules, escalation thresholds, response policies and briefing format.

Skills make performance more consistent. They prevent the agent from reinventing the process each time.

3.4 Tools make action possible

Tools are concrete operations: search news, retrieve prices, read an email thread, send a reply, check a calendar, search contacts, fetch a document.

Without tools, the agent can reason and write, but it cannot reliably interact with the outside world.

3.5 MCPs provide the standard connection layer

MCP is best understood as a standardized way for a host or agentic application to discover and invoke capabilities supplied by servers. The server advertises tools and schemas, receives structured invocations and returns structured results [2].

The MCP server may wrap an API, a database, a local filesystem, an enterprise application or a custom program. It can also include validation, authentication, logging and transformation.

It is not necessarily intelligent. Intelligence may exist behind it, but the protocol itself does not require another model.

3.6 Applications execute and persist

Yahoo Finance supplies market data. Gmail stores and transmits email. Google Calendar or Exchange stores events. A contact database stores relationships. A document repository stores institutional knowledge.

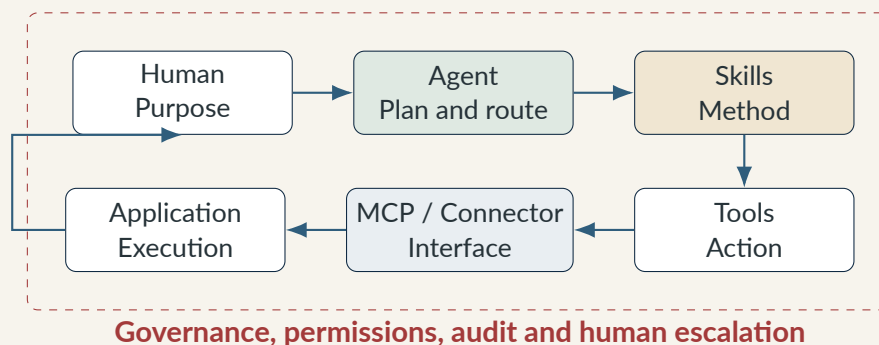
These systems do not need to understand the full human objective. They execute narrow operations.

3.7 Governance determines legitimate autonomy

The difference between an impressive demonstration and a trustworthy production system lies in governance.

The system should know which sources are approved, which actions are reversible, which communications may be sent automatically, which data are confidential, which decisions require a human and how every action will be recorded.

Governance transforms capability into controlled institutional behavior. This emphasis on governance, accountability and continuous risk management is consistent with the NIST AI Risk Management Framework [3].



4 Conclusion: The City as the Better Metaphor

The dominant image of artificial intelligence is still the solitary brain: a single model receiving a question and producing an answer. That image is increasingly inadequate.

The more consequential systems resemble organizations. They combine reasoning with operational methods, specialized tools, secure interfaces, external applications and governance rules. Their intelligence is not located in one component alone. It arises from coordinated specialization.

The Research City shows how a broad analytical objective becomes a plan, how the plan becomes a series of structured requests, how MCP-enabled services connect the agent to diverse providers, and how evidence is transformed into a coherent report.

The Executive City shows a more advanced frontier. The system not only perceives and explains; it acts. It handles routine communications, coordinates across applications

and preserves the scarce attention of the chief executive. Yet it also knows when to stop and return authority to the human.

That last capacity may be the most important of all.

The purpose of an autonomous system is not to remove the human from every decision. It is to construct an intelligent division of labor: machines handle scale, repetition, retrieval and coordination; humans retain purpose, accountability, values and judgment.

In both cities, the visible outcome is simple. A report appears. A morning briefing arrives. But the simplicity is achieved through architecture.

Agents think. Skills discipline. Tools act. MCPs connect. Applications execute. Governance protects. Humans lead.

And this is the real promise of autonomous systems: not a machine that replaces the city, but a new city organized around the intelligent amplification of human purpose.

Final reflection

The machine should not inherit the chief executive's authority merely because it has inherited access to the chief executive's applications. True autonomy is not unlimited action. It is competent action within a deliberately governed mandate.

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End of Paper

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